

Disturbance Aware Dynamic Power Reduction in Synchronous 2RW Dualport 8T SRAM by Self-Adjusting Wordline Pulse Timing

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Abstract— An effective design is proposed to reduce dynamic power consumption for a common clock synchronous 2-read/write (2RW) dual-port (DP) 8T SRAM. A self-adjusting wordline (WL) pulse timing control circuit is newly introduced for read/write operations. Row address inputs of port A and port B are compared in each cycle to detect the same row access or not. In the same row access from both ports, the disturbance should happen, which is an inherent mode of 2RW DP 8T SRAM. Then the WL pulse width is extended to prevent the disturbance, while maintaining sufficient read/write margins. In the different row access, where there is no disturbance, the WL pulse width is shortened to reduce excessive bitline discharge power. Test chips are designed and fabricated to implement the proposed 2RW DP SRAM macros on 40-nm, 28-nm, and 7-nm Fin-FET technologies. Measured data show that read and write powers are reduced respectively by 6–13% and 13–28% with the proposed circuits. No speed degradation is found compared to conventional designs. Area overheads are found to be less than 1%.

Index Terms—7-nm, 8T, 28-nm, 40-nm, Disturbance, Dual-port, Dynamic power, row access, SRAM, Wordline pulse, $V_{\min}$

I. INTRODUCTION

ALONG with scaling down of devices, the total memory capacity embedded in a die tends to increase year by year.

To achieve much energy-efficient computing execution while maintaining high performance, parallel-processing systems with many cores are significantly demanded [1]–[8] combining with power control schemes like a power-gating or a dynamically voltage-frequency scaling (DVFS). In such green high-performance energy-efficient computing systems-on-a-chip (SoCs), embedded multi-port SRAMs are frequently required as well as single-port SRAMs. Because the multi-port SRAMs role as register files with many ports, cache/buffer memories to enable to access the data in parallel between processing units and memories.

Typical multi-port cache/buffer memories in parallel-processing systems have mainly two individual ports. A 1-read/1-write (1R1W) 2-port (2P) SRAM with decoupled read BL 8T bitcell is widely used for a buffer memory of imaging processing or cache of multi-core CPUs [9]–[20]. However, these 1R1W 2P 8T bitcells have the read-only access functionality on port B. It is not able to write operation from

both ports which are often required in the parallel computing system. High-density pseudo-DP SRAMs with single-port 6T SRAM bitcell were also proposed [21], [22], but those have speed shortcomings because of double-pumping of the internal clock.

Meanwhile, 2-read/write (2RW) dual-port (DP) SRAMs are used specially as a data cache or shared cache memory for multi-core CPUs architectures [1]–[4]. DP SRAMs also function as block RAMs in FPGAs [5], [6] or buffer memories for reconfigurable processors [7], [8]. In the hardware implementation of 2RW DP SRAMs, the differential 2RW DP 8T bitcell is necessary to support read operations and write operations for both port A and port B for high-speed operation.

Fig. 1 presents a schematic diagram of a 2RW DP 8T SRAM bitcell. It has two pairs of wordlines (WLs) and bitlines (BLs) as WL_A , WL_B , BL_{A_T} , BL_{B_T} , BL_{A_B} , and BL_{B_B} for simultaneous read operations and/or write operations in parallel. Two pull-up PMOS (PU1, PU2) and two pull-down NMOS (PD1, PD2) make cross-coupled inverters as a latch with complementary internal nodes (MT, MB). There are two pairs of NMOS pass-gates (PG1_A, PG2_A and PG1_B, PG2_B) to access from both ports. Fig. 1 also depicts a simple block diagram of a common clock synchronous 2RW DP SRAM macro with an 8T bitcell. RA_A (RA_B), CA_A (CA_B) and DA/Q_A (DB/Q_B), respectively denote row address inputs, column address inputs and data inputs/outputs for port A (port B). CEN_A (CEN_B) and WEN_A (WEN_B) respectively stand for chip enable and write enable signals. The signal “CLK” stands for the common clock for port A and port B. The power and ground symbols shown in Fig. 1 are commonly used for all later figures. Fig. 2 presents a timing chart of a 2RW DP 8T SRAM macro. Both of port A and port B are synchronously operated with rising CLK edge. The control signals of the chip enable and write enable are independent for each port. If the CEN_A (CEN_B) is “1”, the port works as no operation (NOP). If the CEN_A (CEN_B) is “0” and the WEN_A (WEN_B) is “1”, the port works as read operation, otherwise it works as write operation when the WEN_A (WEN_B) is “0”. Each port has its own address inputs, data inputs/outputs, enabling to operate NOP, read or write operation independently.

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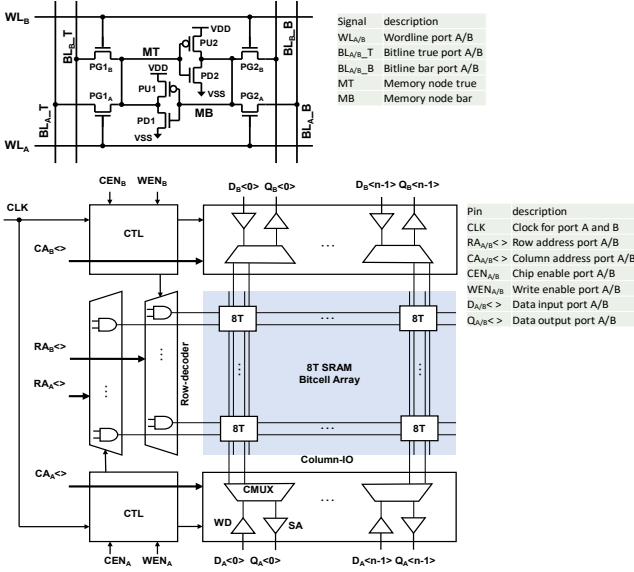


Fig. 1. Schematic diagram of 2-read/write (2RW) dual-port (DP) 8T SRAM bitcell and block diagram of common clock synchronous 2RW DP SRAM macro.

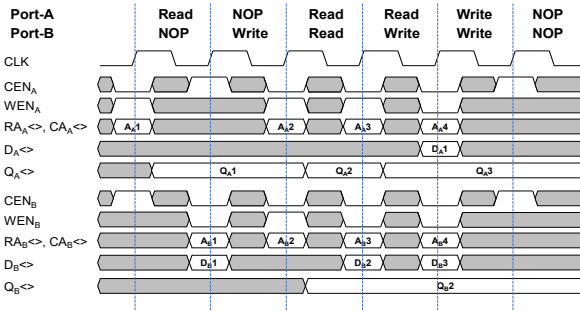


Fig. 2. Timing chart of common clock synchronous 2RW DP SRAM.

Many reports of the relevant literature describe studies related to 2RW 8T DP SRAM designs[23]–[40], [47], [48]. Earlier works [24], [25] have particularly addressed enhancement of write/read margins for lower-voltage operation. Compact 2RW 8T bitcells were proposed to improve the macro density [23], [27]. Prevention or detection of disturbance issues between both port accesses have been discussed in earlier reports of the literature [27]–[31], [39], [46]. Demonstrations on FD-SOI devices [36]–[37] and on Fin-FET devices [34], [35], [39] have also been reported. A computing-in-memory based on the 2RW 8T SRAM bitcell has also been proposed recently as another application [40]. Several works have been undertaken for reducing the standby power of 2RW DP SRAM [26], [27], [37], but few studies have particularly addressed dynamic power reduction of 2RW DP SRAMs [23], [38]. To achieve an energy-efficient parallel computing system, dynamic power reduction is important along with leakage power reduction.

As described in this paper, we propose a dynamic power reduction scheme for common clock synchronous 2RW DP SRAMs under both read operation and write operation. By detecting the same and different row address access, WL pulse timing is self-adjusted to minimize the BL discharging power

while maintaining sufficient read and write margins [38]. We confirmed the proposed technique is effective on a 7-nm Fin-FET advanced technology as well as 28-nm and 40-nm planar bulk CMOS technologies.

The organization of this paper is the following. Section II describes the disturbance issues in DP SRAM design. Section III describes the concept of reducing dynamic power in DP SRAM macros. In Section IV, a proposed circuit aware of disturbance issues to reduce dynamic power is presented. In Section V, designed and fabricated test chips on 40-nm, 28-nm, and 7-nm Fin-FET technologies with well-balanced 8T bitcells are demonstrated. The given silicon measurement results are shown. A brief conclusion is presented in Section VI.

II. DISTURBANCES IN 2RW DUAL PORT 8T SRAM

A. Read Disturbance

The DP 8T SRAM has two disturbance modes: read-disturbance and write-disturbance. Fig. 3 presents the read-disturbance at the same row address access where both WL_A and WL_B are activated. A different row access mode, as shown in Fig. 3(a), works as a single-port (SP) 6T SRAM, activated either WL_A or WL_B, so disturbance from the other port never happens. In the same row, access mode shown in Fig. 3(b) undesirable cell current flows through the other BL into the pull-down (PD) NMOS in the accessed bitcell. In that case, the read current (I_{read}) of the target BL through the pass-gate (PG) and PD NMOSs decreases by additional cell current from the other BL. As a result, the discharge speed of BL_A decreases. Here, the current disturbs another port operation, defined as a “disturbing current” (I_{dist}), as shown in Fig. 3(b). The I_{dist} is come from pre-charge circuit in another port. In this work, the technique of pre-charging unselected columns has been used to suppress the peak rash current in pre-charging BLs. In other words, pre-charge PMOSs in unselected columns are always on.

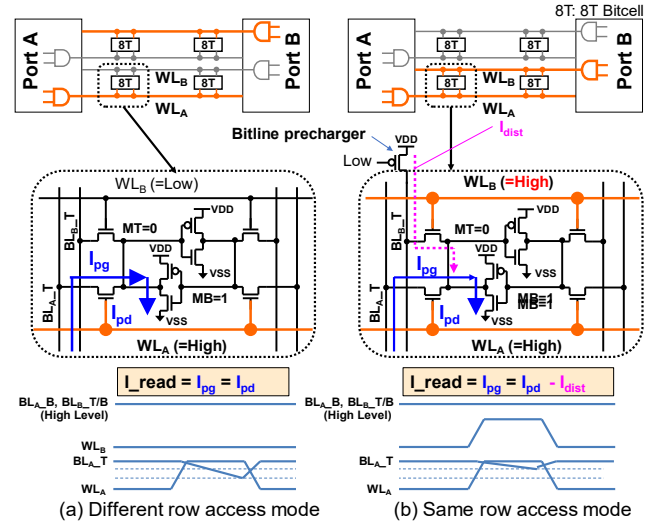


Fig. 3. Read disturbance issue.

B. Write Disturbance

Fig. 4 shows the other disturbance issue in the write operation. It is also disturbed by BL current as well as a read-disturbance

issue. In a different row access mode as shown in Fig. 4(a), the write operation is performed by competition between pulling down current I_{PG} by $PG1_A$ in Fig. 1 and pulling up current I_{PU} by $PU1$ in Fig. 1. Whereas, in the same row access mode as shown in Fig. 4(b), I_{dist} by the pre-charge circuit though $PG1_B$ is added as one of pulling up current. In this case, the write ability to flip the stored data is weaker than the different row access mode.

Furthermore, even if the voltage of the MT node would become lower and $PU1$ would turn off, the MT node does not completely become 0 V. I_{dist} remains as long as WL_B is activated; the MT node is raised from the 0 V level by competition between $PG1_B$ pulling up and $PG1_A$ pulling down. Then, the MT node becomes a certain intermediate voltage. If $PU2$ is weak and if the pulse width of WL_A is not sufficiently wide, then the voltage difference between MT and MB might not be sufficient to flip while WL_A is activated. MT is lowered when WL_A reaches a low level. As a result, MT and MB nodes are not flipped, resulting in the write fail.

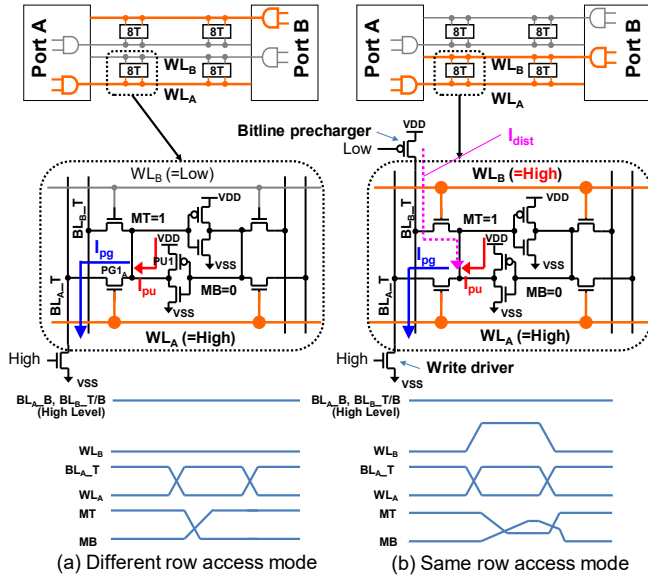


Fig. 4. Write disturbance issue.

C. Theoretical Consideration

This paragraph presents theoretical discussion of the degradation of read cell current (I_{read}) of the DP 8T bitcell by read disturbance. Fig. 5 shows the current flows of two PG NMOSs and PD NMOS in the bitcell at different row access and same row access where the datum stored in the bitcell is “0”. The read current of BLA in the different row access, which is indicated as $I_{readA-diff}$, flows to VSS through corresponding PG and PD of the bitcell. In the different row access mode, each current through PG_A (I_{PGA}) and PD (I_{PD}) is the same as $I_{readA-diff}$ shown as Eq. (1). Each I_{PGA} and I_{PD} is calculated using fundamental NMOS current equations for the linear and saturation region, as presented respectively in Eqs. (2) and (3) [41].

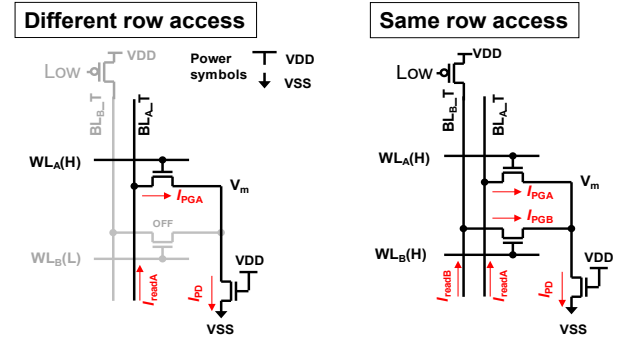


Fig. 5. Current analysis in different and same row accesses.

$$I_{readA-diff} = I_{PGA} = I_{PD} \quad (1)$$

$$I_{PGA} = \frac{1}{2} \mu C_{ox} \frac{W_G}{L_G} (V_{DD} - V_{th-PGA} - dV_{th-PGA} - V_m)^2 \quad (2)$$

$$I_{PD} = \mu C_{ox} \frac{W_D}{L_D} ((V_{DD} - V_{th-PD})V_m - V_m^2) \quad (3)$$

Here, V_m denotes the voltage of complementary internal node of bitcell; V_{DD} is the supply voltage. Also, V_{th-PGA} and V_{th-PD} respectively represent threshold voltage of PG_A and PD. μ stands for the mobility and C_{ox} denotes the gate capacitance. W_G , L_G , W_D and L_D respectively represent the gate width and length of PG NMOSs and PD NMOS. Precisely, the threshold voltage of PG_A is affected by body bias effects, the fluctuation of threshold voltage (dV_{th-PGA}) is defined as (4). Variables ϵ , N_A , and ϕ_F respectively represent permittivity, Boltzmann's constant, and Fermi level.

$$dV_{th-PGA} = \frac{\sqrt{2\epsilon q N_A}}{C_{ox}} (\sqrt{V_m + 2\phi_F} - \sqrt{2\phi_F}) \quad (4)$$

To simplify the equation, the body bias effect is omitted and the $1/2\mu C_{ox}(W_G/L_G)$ and $1/2\mu C_{ox}(W_D/L_D)$ are replaced to gain factor β_G and β_D , and V_{th-PGA} and V_{th-PD} are defined as the same NMOS threshold voltage (V_{thn}). Then (2) and (3) are solved for $I_{readA-diff}$ as (5). Similarly to different row access, $I_{readA-same}$ is given as (6). By comparing the Eqs. (5) and (6), it is delivered that the $I_{readA-same}$ is smaller than $I_{readA-diff}$ (7).

$$I_{readA-diff} = \frac{1}{2} \beta_G \left(1 - \frac{1}{1 + \frac{\beta_D}{\beta_G}} \right) (V_{DD} - V_{thn})^2 \quad (5)$$

$$I_{readA-same} = \frac{1}{2} \beta_G \left(1 - \frac{1}{1 + 2\frac{\beta_D}{\beta_G}} \right) (V_{DD} - V_{thn})^2 \quad (6)$$

$$\therefore I_{readA-same} < I_{readA-diff} \quad (7)$$

D. Practical Simulation Results

To ascertain the practical effects of read-disturbance and write-disturbance, Monte Carlo simulations for 40-nm, 28-nm, and 7-nm technologies are conducted with considering high-sigma local variations. Fig. 6 shows the reading and writing waveforms of WL and BL in the same/different row access modes on a 28-nm DP 8T bitcell at the slow-NMOS and slow-PMOS (SS) corner, low voltage, and 40°C condition. Each

waveform is given by 1000 iterations of Monte Carlo simulation with simply accelerating 2 x local variation of each transistor. The worst corner waveforms correspond approximately to 6σ , showing the range of variations, but not practical distributions. In the write operation, as presented in Fig. 6, the negative BL write-assist technique is introduced to 28-nm and 7-nm in this work, so that the BL is biased to negative voltage.

Fig. 7 defines the read-operation and write-operation margins. ΔV_{sense} is the differential voltage between BL_{A_T} and BL_{A_B} (ΔV_{BL}) for the sense amplifier (SA) to sense the read data correctly, and T_{sense} is the duration time that ΔV_{BL} reaches required ΔV_{sense} after WL_A is activated (Fig. 7(a)). For the write operation, ΔV_{flip} is the required voltage difference to flip the complementary nodes of MT and MB (ΔV_{MEM}) before WL_A lowers. T_{flip} is the duration time that ΔV_{MEM} reaches required ΔV_{flip} after WL_A and BL_T are activated (Fig. 7(b)). T_{sense} and T_{flip} of the same row access have longer durations than those of different row access, as shown in Fig. 6.

TABLE I presents simulation results of the practical T_{sense} and T_{flip} duration times in 40-nm, 28-nm, and 7-nm technologies. T_{sense} and T_{flip} in the same row access mode are worsened by approx. 39% and 145% compared to the different row access mode. Each value in TABLE I is obtained by accurate 6σ worst-case simulation which is obtained using importance sampling Monte Carlo method [42]. Note that the T_{flip} of 28-nm is larger than 40-nm because the 28-nm is optimized for eFlash technology that is minimized the leakage power but not for speed.

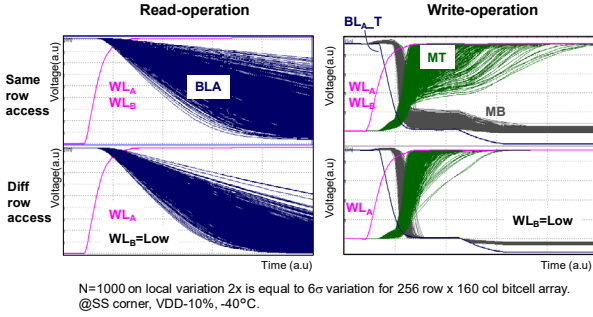


Fig. 6. Monte-Carlo simulation waveforms on 28-nm for read operation and write operation.

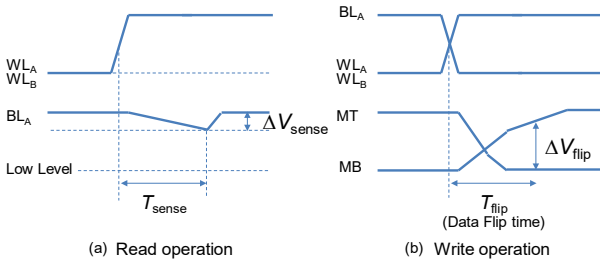


Fig. 7. Definition of ΔV_{sense} and T_{sense} in read operation and ΔV_{flip} and T_{flip} in write operation.

TABLE I

SUMMARY OF T_{sense} AND T_{flip} IN SAME/DIFF ROW ACCESS MODE
Condition: SS, $V_{\text{typ}} - 10\%$, -40°C

Node	$T_{\text{sense}} @ -6\sigma$ ($\Delta V_{\text{sense}} = 50 \text{ mV}$)			$T_{\text{flip}} @ -6\sigma$ ($\Delta V_{\text{flip}} = V_{\text{typ}} \times 0.9 \times 0.7$)		
	Same (ps)	Diff (ps)	Same/Diff	Same (ps)	Diff (ps)	Same/Diff
40-nm	754	618	122%	1253	526	238%
28-nm	477	344	139%	1560	636	245%
7-nm	254	225	113%	1072	452	237%

III. DYNAMIC POWER REDUCTION

In the preceding section II, we explained that read-disturbance and write-disturbance respectively induce the degradation of duration times of T_{sense} and T_{flip} . Those disturbances should be regarded as designing the internal timing of the 2RW DP SRAM macro for stable operation so that it is usually set of the worst timings at the same row access condition irrespective any access use cases. However, it has excessive timing margins at the single-port access either by port A or port B and difference row address access by both ports. It induces undesirable power overhead as discussed in the following. In other words, if the internal timing can be adjusted dynamically along with any access modes, then it has opportunities to reduce excessive power consumption.

Fig. 8 presents waveforms of T_{sense} and T_{flip} in the same and different row access modes by adjusting the WL pulse width to optimize the timing and power consumption. Typically, the WL pulse width for the read operation is shorter than the write operation because of poor write recovery in the same row access [35]. Theoretically, BL charging and discharging powers are delivered by the following equations. P_{read} and P_{write} , which are read and write power per I/O bit, are defined respectively as (8) and (9).

$$P_{\text{read}} = V_{\text{DD}} \times \left(\int_{t_0}^{t_R} I_{\text{read}} dt \right) \times N_{\text{M}} \quad (8)$$

$$P_{\text{write}} = V_{\text{DD}} \times \left(\int_{t_0}^{t_W} I_{\text{read}} dt \right) \times (N_{\text{M}} - 1) + \frac{1}{2} C_{\text{B}} V_{\text{DD}}^2 \quad (9)$$

Here, C_{B} stands for BL capacitance, V_{DD} denotes supply voltage, N_{MUX} expresses column number per I/O, t_0 signifies WL assert timing, and t_R and t_W respectively mean WL negate timings in read operation and write operation. Fig. 9 shows the related dynamic power consumption of the 2RW DP SRAM macro. It mainly consumes power for charging and discharging BLs during activated WL. In the read operation, it is necessary to discharge the selected column BLs by ΔV_{sense} , and unselected column bitcells consume as same as selected column BLs. In write operation, the selected column BLs are fully discharged by the write-driver circuit, and unselected column BLs consume power similarly to the read operation. If the WL pulse width in the different row access is the same as that of in the same row access, then BLs consume excessive power, as shown in Fig. 9. Furthermore, the technique of pre-charging unselected columns has been usually used to suppress the peak rash current in pre-charging BLs, as shown in Fig. 5. In that case, unselected BL

might consume more power than fully discharged BL for write operations.

Here, we assume the probability of occurrence of the same row access by both ports is $1/\text{row}$ if the row address is accessed randomly. For example, it is only 0.4% in the 256-row macro configuration. Although a small probability of the same row access exists, the same row access should be regarded as a worst condition for maintaining the design margin. In other words, 99.6% operation has excessive timing margins because the different row access has better timing margins for both read operation and write operation, as shown in Fig. 10. These overtuning margins consume undesirable dynamic power by extra discharge of BL capacitances.

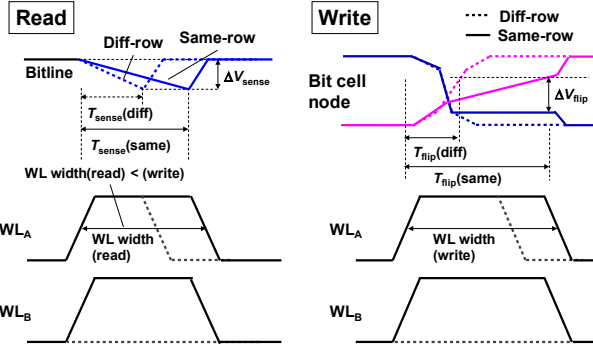


Fig. 8. Waveform of same/diff row access and inflection of $T_{\text{sense}}/T_{\text{flip}}$.

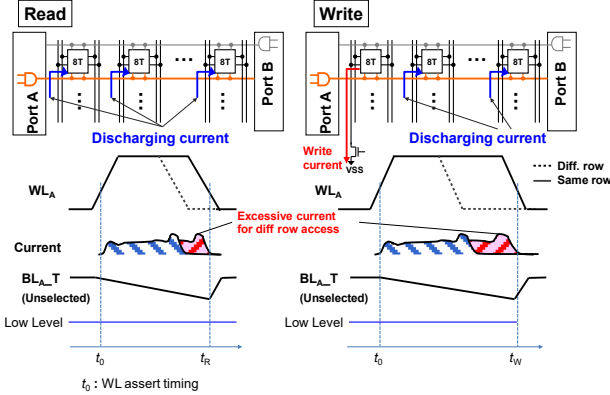


Fig. 9. Power consumption of SRAM macro by discharging BLs.

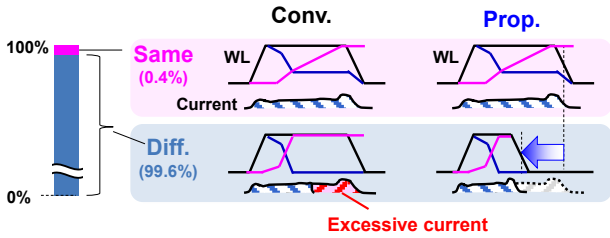
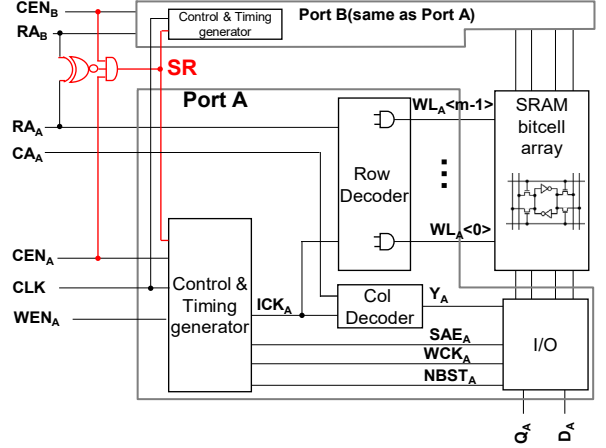


Fig. 10. Probability of same/diff. row access and power reduction by reducing WL pulse width in write operation.

IV. ADJUSTING WL PULSE FOR POWER REDUCTION

To eliminate the extra dynamic power consumption, we propose a self-adjusting WL pulse timing control circuit. Fig. 11 presents a block diagram of the proposed circuit, where RA_A and RA_B respectively represent row address inputs of port A

and port B. The input signal CLK is a common clock for port A and port B as a synchronous operation between both ports. If both row address inputs of port A and port B are exactly the same and both enable signals of port A (CEN_A) and port B (CEN_B) are “0” (active-low), then the row address comparator (XNOR) signal SR becomes “1” to set the flag as the same row access mode. Otherwise, the SR signal becomes “0” when both row addresses are not the same or either CEN_A or CEN_B is “1” (not enabled). Fig. 11 also shows a truth table for each operation mode over each row-address input (RA_A , RA_B) and each enable signal (CEN_A , CEN_B).



RA_A , RA_B	CEN_A	CEN_B	SR	Mode
$RA_A = RA_B$	0	0	1	Same row access
$RA_A \neq RA_B$	0	0	0	Different row access
X	0	1	0	Port A operation
X	1	0	0	Port B operation
X	1	1	0	No operation

Fig. 11. Block diagram of proposed DP SRAM and truth table for each operation mode over each row-address input (RA_A , RA_B) and each enable signal (CEN_A , CEN_B).

The SR flag is provided to following row-decoder and column IO blocks through control (CTL) block to adjust the WL pulse width and related column IO timing signals such as SA enable, WD enable, and pre-charge control. Fig. 12 shows details of the timing generator in the CTL block and column I/O circuit. The timing generator has replica BL, shown as RBL_A in Fig. 12, which can determine WL pulse width. The replica BL is connected to two kinds of replica bitcells. Replica on-bitcells discharge the replica BL to generate the appropriate timing, and another replica off-bitcells are parasitic capacitance load imitating bitcell capacitance. The proposed circuit has further parasitic capacitance C_{SR} , which is composed by NMOSs, connected to node RBL_A shown in Fig. 12. If the signal SR is “1”, then the parasitic capacitance C_{SR} becomes effective and increases the propagation delay. As a result, the WL pulse width increases by increased RBL_A delay. The triggered timing of the sense-amplifier enable (SAE) is also pushed out in the read operation at same row access mode (SR=“1”). In the write operation, further offset delay is added to the path through multiplexer to ensure the write ability by further extension of the WL pulse width at the same row access mode. As a result, the write-driver period is also increased at the same row address

mode. The inserted logic delay is composed by inverter chain with logic MOSs optimized at the worst PVT corner (NMOS: Slow - PMOS: Slow, low voltage and low temperature). It has small excessive delays at the other PVT corners, but it should be needed to keep the write margin for all PVT conditions. If the inserted delay is generated by SRAM bitcell circuits, it has better sensitivity against PVT variations. However, generating the delay by SRAM bitcells needs an additional memory cell array area. The logic delay composed by logic MOSs has contrary almost no area impact by implementing in the peripheral control block.

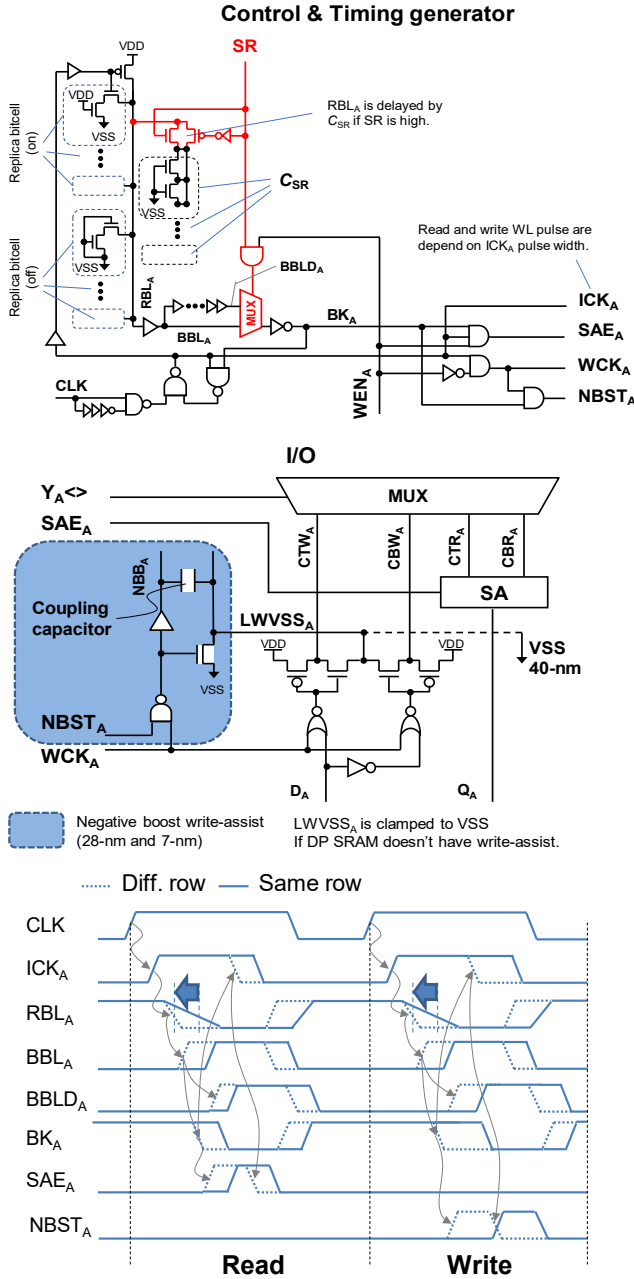


Fig. 12. Proposed self-adjusting WL pulse timing control circuit and I/O circuit.

In demonstrations on 28-nm and 7-nm, the write-assist techniques are applied to compensate poor write margin of the 2RW DP 8T SRAM bitcells at the same row access. Many

write-assist techniques have been reported to date [24], [25], [30], [32], [34], [35]. In this work, the negative BL write-assist techniques [24], [25], [32] are applied 28-nm and 7-nm designs not by 40-nm design, as presented in Fig. 12. Although the negative boost consumes further dynamic power because it drives coupling capacitors, it is necessary irrespective of the same or different row access in the advanced processes. The proposed circuit has no penalty for access time or address setup time because the setup clock path is longer than data path. The area overhead is less than 1%.

Fig. 13 shows SPICE simulation waveforms of the 28-nm DP SRAM macro at the worst process-voltage temperature (PVT) conditions in the read operation and write operation. The simulated WL pulse width for the read operation can be shortened by 25% in the different row access compared to the same row access. Here, we assume that the same BL swing: delta voltage between CBR_A and CTR_A in Fig. 12 is almost identical to the different row access. However, the simulated WL pulse width for the write operation, as presented in Fig. 13(b), can be shortened further by 51% for different row access compared to the same row access. This simulation demonstrates that the write-disturbance strongly affects the differences of the WL width between the same and different row access. From the power reduction simulation results presented in Fig. 14, we estimated a dynamic power comparison in the two types of column MUX (MUX8 and MUX2) configurations in 40-nm, MUX4 type in 28-nm and MUX2 type in 7-nm under the worst PVT conditions.

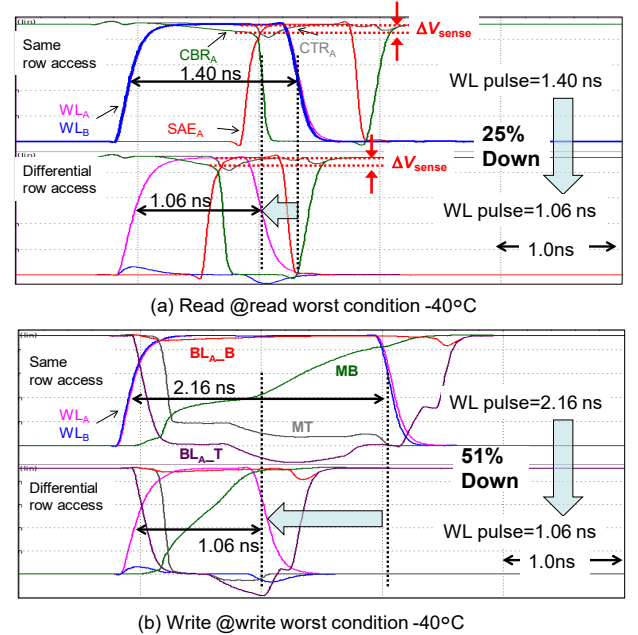


Fig. 13. Simulation waveforms of 28-nm DP SRAM 48-kbit MUX4 (1024-word x 48-bit) with adjusting WL pulse timing control.

Fig. 14 portrays the estimated results of charging/discharging per column by introducing proposed circuits. Dynamic power used for read operation and write operation is reduced in the respective cases by 13–29% and 18–56%. The power consumption of the selected column in the write operation becomes smaller than unselected column if the technology node becomes smaller in Fig. 14(b). It shows that T_{flip} becomes larger

in an advanced technology and it requires longer WL pulse width to perform the write operation. Fig. 15 portrays the PVT dependency of dynamic power reduction by proposed circuit in 40-nm MUX8. Read power reduction shows almost same rate in each PVT. It means WL pulse width increased by C_{SR} well matches with discharging current of the bitcell in reading. Meanwhile, write power reduction is fluctuated in the different PVT corners. In this work, the logic delay of $BBLD_A$ in Fig. 12 is further added to ensure the T_{flip} degradation. Because the logic delay has the advantage of small area overhead while its PVT dependency does not match with bitcell discharging current. Though the PVT dependency is fluctuated, write power reductions are confirmed in the different PVT corners.

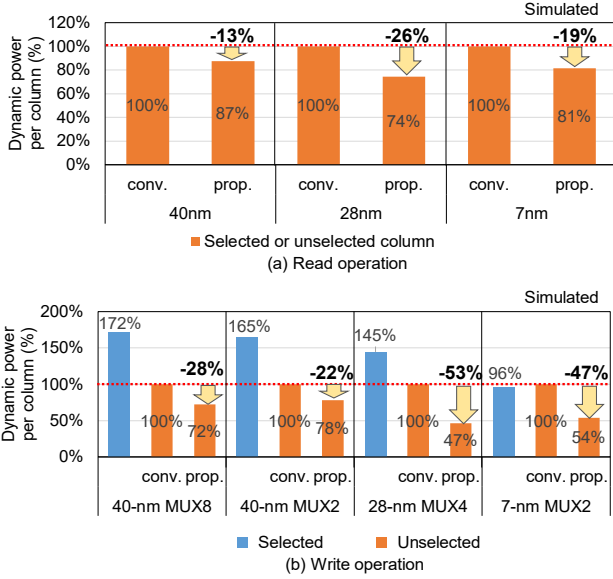


Fig. 14. Simulation result of dynamic power consumption at typical PVT condition.

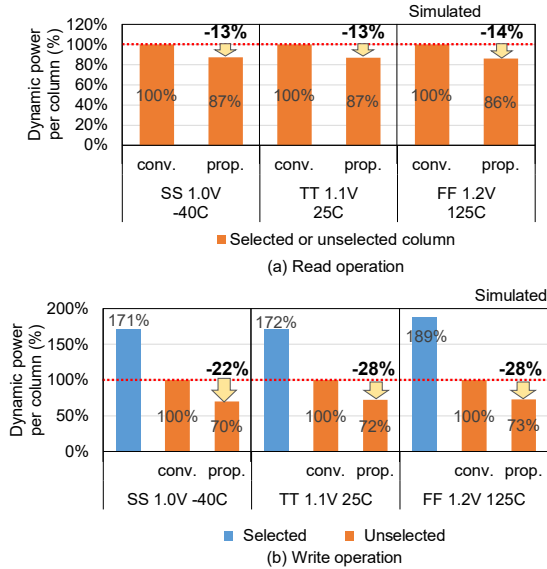


Fig. 15. PVT dependency of estimated dynamic power reduction of 40-nm MUX8.

V. TEST CHIP DESIGN AND EVALUATION

A. 2RW DP 8T SRAM bitcell layout

Layouts of various types for 2RW DP 8T SRAM bitcells have been reported [23],[38],[44] which have lithographic and logic-process-friendly compact layouts with no additional components. Fig. 16 shows representative bitcell layouts. Type (a) [23] have compact area, but not good symmetry between each true/bar BL pair, or between port A and port B because of process variations in manufacturing such as misalignment of photomasks, layout dependent effect, or mismatch of gate length/width by limitations of optical proximity correction lithography. In this work, types (b) [38], [43] and (c) [44] are used for demonstrations because layouts of these types can improve such imbalances. The symmetry of type (b) of DP 8T bitcell layout is validated by the isolated test structure for direct measurement of each I_{read} in 40-nm technology. Fig. 17 presents the measured distributions of I_{read} for A-true, A-bar, B-true, and B-bar BLs implemented using the 40-nm technology. In the graph, all values of I_{read} are normalized by median and sigma. Type (b) has small differences among all I_{read} , whereas the type (a) has offsets between true/bar or port A/B BLs. From the measured I_{read} distributions, the write-margin (WM), read-margin (RM), and static-noise margin (SNM) of the adopted type (b) bitcell is expected to be improved by virtue of the small offsets, resulting in a good minimum operating voltage (V_{min}) and shorter T_{sense} time (access time). The offsets in the type (a) layout should lead to extra design margins, inducing the overhead of power and timing, and deterioration of V_{min} . The type (c) also keeps symmetry, which is used for 28-nm and 7-nm demonstrations in this work. It is expected that type (c) also has small offsets between port A/B and true/bar. In the following discussion, we simulated or measured SRAM macros using the bitcell layouts of Fig. 16(b) for 40-nm and Fig. 16(c) for 28-nm and 7-nm. The coupling noise between BLs in each port is sufficiently small because of the shielded wiring of either the power supply or the ground between each port.

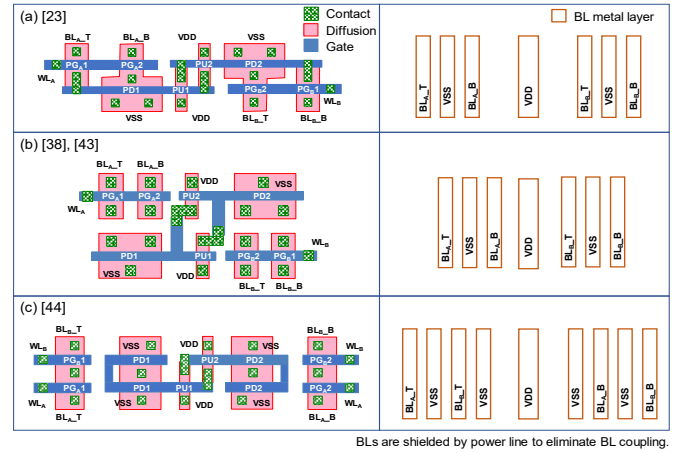


Fig. 16. Layout plots of 2RW DP 8T SRAM bitcells.

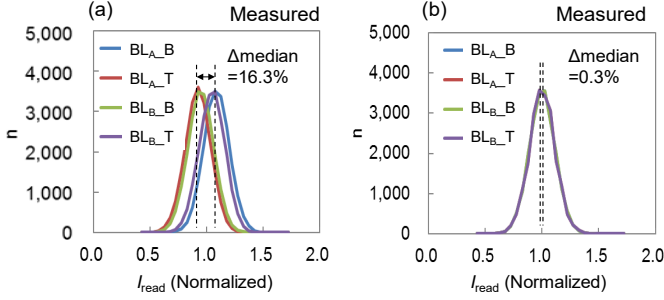


Fig. 17. Measured I_{read} distributions of A/B-true/bar BLs in 40-nm technology ($n=20320$).

B. Test chip implementations and evaluations

Fig. 18 shows die photographs of test chips using 40-nm, 28-nm embedded flash (eFlash) [45], and 7-nm Fin-FET CMOS technologies. Two types of 40-nm DP SRAM macros (MUX2 and MUX8), 28-nm eFlash MUX4 macro and 7-nm FinFET MUX2 macro are implemented in the test chips. Fig. 19(a)–19(d) show cumulative distribution functions (CDFs) of V_{min} for each macro at -40°C worst condition. The median of V_{min} for TT process are 0.71 V (40-nm MUX2 and MUX8), 0.79 V (28-nm eFlash), and 0.54 V (7-nm FinFET). The 28-nm eFlash process is optimized for eFlash IPs. The standby leakage for automotive applications is minimized, so that the threshold voltages of core transistors and SRAM ones are higher than those of the normal CMOS technologies. For that reason, the 28-nm V_{min} is not better than 40-nm and 7-nm.

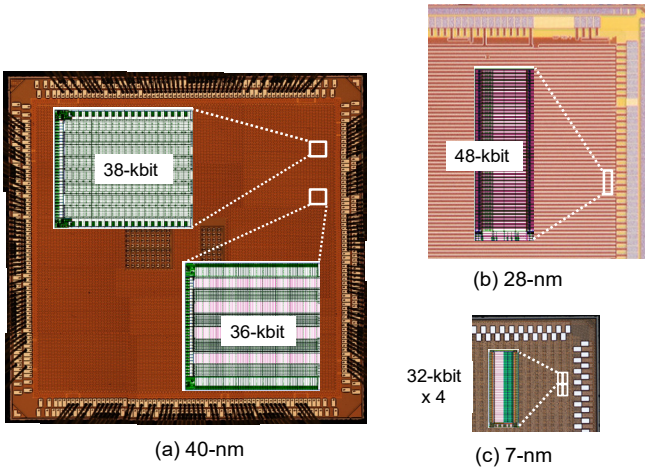


Fig. 18. Photographs of the test chips and layout plots of the proposed 2RW DP SRAM macros: (a) 40-nm [38], (b) 28-nm eFlash [45], and (c) 7-nm FinFET.

Fig. 20 portrays a typical shmoo plot of a 7-nm FinFET DP SRAM macro showing supply voltage vs. read access time. The measured read access time is 0.58 ns at 0.75 V typical supply voltage. Fig. 21 shows the measured dynamic power consumptions of write and read operations vs. the supply voltage. The WL pulse width has been optimized to ensure the read/write margins under the worst condition (for example SS/VDD-10%/-40°C). Measured typical dynamic power consumptions are 19.5 $\mu\text{W}/\text{MHz}$ (40-nm MUX8 read operation), 19.6 $\mu\text{W}/\text{MHz}$ (40-nm MUX8 write operation), 6.9

$\mu\text{W}/\text{MHz}$ (28-nm eFlash read operation), 9.3 $\mu\text{W}/\text{MHz}$ (28-nm eFlash write operation), 3.4 $\mu\text{W}/\text{MHz}$ (7-nm FinFET read operation), and 5.9 $\mu\text{W}/\text{MHz}$ (7-nm FinFET write operation). Here, all powers are obtained under random address accessing. The probability of same row access is less than 0.4%. The measured data show that the dynamic powers of the proposed DP SRAM macros are reduced by 7%, 13%, and 6% for the read operation, and reduced respectively by 18%, 28% and 13% for the write operation compared to the conventional macros for 40-nm, 28-nm eFlash, 7-nm FinFET technologies. TABLE II presents features of the demonstrated test chips. Fig. 22 plots the power comparison over process technologies. The proposed circuit decreases the power consumption as shown at 40-nm. Furthermore, as the process technology evolution and the power supply lowering, the power consumption is decreasing. TABLE III shows the comparison results with other 2RW 8T DP SRAM works in the last 10 years. The dynamic power of this work is smaller than other disclosed work in 28 nm.

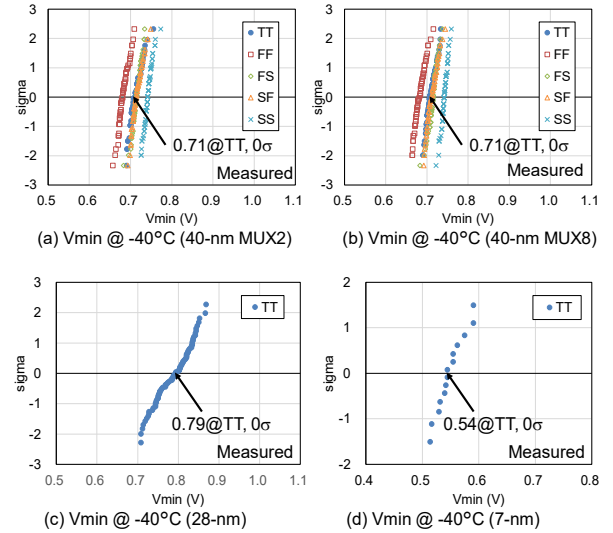


Fig. 19. Distributions of measured V_{min} at -40°C worst temperature.

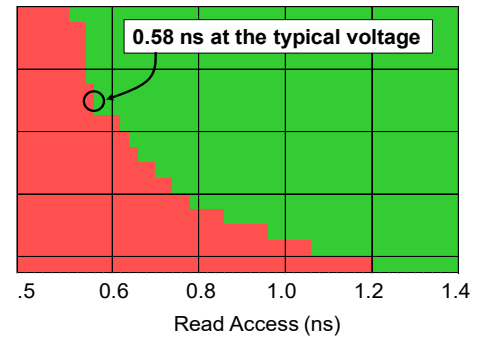


Fig. 20. Shmoo plot of 7-nm 32k-bit DP SRAM macro (supply voltage vs. read access time) at 25°C .

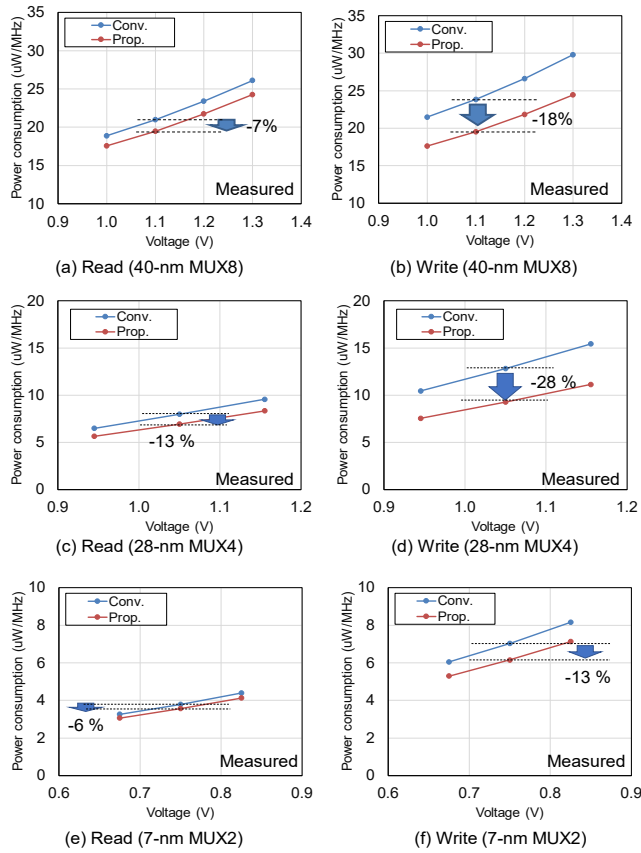


Fig. 21. Measured dynamic power vs. supply voltage at 25°C, TT-process.

TABLE II
FEATURES OF THE TEST CHIPS

Technology node		40-nm	28-nm	7-nm
Configuration	Capacity	36k	38k	32k
	Bit	73	19	48
	Word	512	2k	1k
	Col. MUX	2	8	4
Physical macro size (μm^2)		195.3 x 191.6	180.9 x 203.7	95.3 x 297.1
Bit density (Mbit/mm ²)		0.99	1.06	1.74
Read access time (ns)	Typical	1.4	1.6	1.54
	Typical	500	438	455
Frequency (MHz)	Typical	0.71	0.71	0.79
	Typical	0.71	0.79	0.54
Measured power ($\mu\text{W/MHz}$)	Read	-	19.5	6.9
	Write	-	19.6	9.3
Energy (pJ/bit/access)	Read	-	1.026	0.144
	Write	-	1.032	0.193

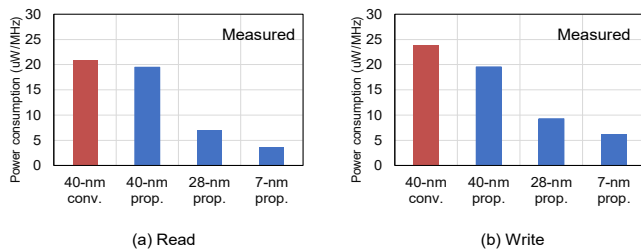


Fig. 22. Dynamic power comparison over 40-nm, 28-nm, and 7-nm technologies.

TABLE III
COMPARISON WITH OTHER 2RW DP SRAM WORKS

	TCAS-II 2012 [30]	SOC 2013 [31]	VLC 2014 [32]	A-SSCC 2014 [33]	IEDM 2015 [34]	S3S 2017 [37]	ISSCC 2019 [39]	ISNCC 2020 [48]	TVLSI 2021 [47]	This work		
Technology (nm)	40	40	28	40	16	65	7	7	40	40	28	7
Capacity (kbit)	1024	72	32	24	64	32	16	512	16	38	48	36
Bit Density (Mbit/mm ²)	-	1.48	2.1	0.86	3.60	0.75	-	-	0.85	1.06	1.74	6.00
Access time (ns)	4.2	-	0.6	1.67	0.69	4.51	-	-	2.32	1.60	1.54	0.58
Frequency (MHz)	-	-	1000	-	-	-	2100	1359	-	435	455	1207
Write power ($\mu\text{W/MHz}$)	-	-	10.4	-	10.3	-	-	-	-	19.6	9.3	5.9
Read power ($\mu\text{W/MHz}$)	-	-	-	-	5.5	6.9	-	-	-	19.5	6.9	3.4
Typical VDD (V)	1.0	1.1	1.0	1.25	0.9	0.75	0.8	-	1.05	1.1	1.05	0.75

VI. CONCLUSION

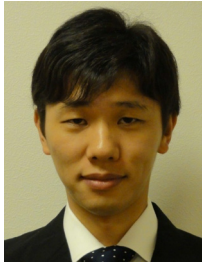
A self-adjusting WL pulse timing control circuit is proposed for a common-clock synchronous 2RW DP 8T SRAM to reduce dynamic power consumption in read/write operations, with awareness of disturbance issues. Test chips are designed and fabricated to implement proposed 2RW DP SRAM macros on 40-nm, 28-nm eFlash, and 7-nm Fin-FET technologies. Measured data show that reading and writing powers are reduced respectively by 6–13% and 13–28% with proposed circuits. No speed degradation was found compared to conventional designs. Area overheads are less than 1%.

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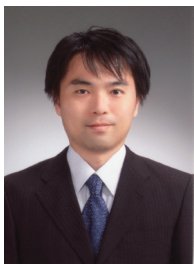


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